CAMPANA STRONG APPROXIMATION ON GENERALIZED MARKOFF SURFACES

DASHENG WEI

ABSTRACT. For every smooth projective generalized Markoff surface X_m, every hyperplane section H, and every integer $n \geq 2$, we establish Campana strong approximation with the Brauer–Manin obstruction for $\big(X_m, (1-1/n)H\big)$.

1. INTRODUCTION

Let k be a number field, fix $a, b, c \in k^\times$, and let $m \in k \setminus \{0, 4abc\}$. Consider the generalized Markoff surface

$$X_m : \quad (ax^2 + by^2 + cz^2)w - xyz = mw^3 \subset \mathbf{P}^3_k. \tag{1.1}$$

Its boundary is the triangle $D = D_x + D_y + D_z$ with

$$D_x = \{x = w = 0\}, \quad D_y = \{y = w = 0\}, \quad D_z = \{z = w = 0\}.$$

Writing $U_m = X_m \setminus D$, we have

$$U_m : \quad ax^2 + by^2 + cz^2 - xyz = m. \tag{1.2}$$

The surface X_m is smooth when $m \notin \{0, 4abc\}$. The fixed coefficients a, b, c are suppressed from the notation X_m and U_m. In the special case $a = b = c = 1$, the geometry and the Brauer group were studied in [CTWX20, LM21].

The classical Markoff equation $x^2 + y^2 + z^2 = 3xyz$ first appeared in 1879 [Mar79]. Its positive integral points carry an action of a discrete group and are generated by $(1, 1, 1)$. Hurwitz studied its higher-dimensional analogues [Hur07]. Much subsequent work explores connections between these equations, counting problems, dynamical systems, and local-global principles. Important results include strong approximation modulo primes [BGS16], asymptotic counting for integral points [GMR19], and relations to Painlevé dynamics on character varieties [CL09].

For the family $x^2 + y^2 + z^2 - xyz = m$, Ghosh and Sarnak [GS22] developed reduction theory for integral solutions. They proved the integral Hasse principle for almost all integer parameters and exhibited infinitely many failures. Loughran and Mitankin [LM21], together with Colliot-Thélène, Xu, and the author [CTWX20], computed the full Brauer group and

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further studied failures of the integral Hasse principle via the Brauer–Manin obstruction. Some twisted and number-field generalizations have been treated in [Che21, Dao24, KSS24].

Campana points [Cam05] interpolate between rational and integral points. For a finite weight n, a positive boundary multiplicity is permitted only when it is at least n. If n is infinite, then Campana points recover S-integral points on the complement. Here we consider finite weights. With the basic notion of Campana points in place, numerous related arithmetic results have been obtained. A Manin-type conjecture and counting results for Campana points are developed in [PSTVA21], with further counting results in [BY21, PS24]. Turning to approximation theory, weak approximation for Campana points has been studied for projective spaces and quadrics in [NS24, MNS24, Wei26]. Failures of local–global principles for Markoff orbifold pairs were studied by Mitankin and Uhlemann [MU25].

We now return to the generalized surface over a number field k. Let Π be a hyperplane in $\mathbf{P}^3$, and let $H = X_m \cap \Pi$ be the corresponding (reduced) hyperplane section. Set $\Delta_H = (1-1/n)H$, where $2 \le n < \infty$. Fix a finite set $S \supseteq \Omega_k^\infty$ and a proper flat integral model $\mathscr{X}_m$ over $\mathcal{O}_S$. In this paper, we follow [NS24, Definitions 2.2–2.4] for the local intersection multiplicity. Let $\mathcal{C}_v(\mathscr{X}_m, \Delta_H)$ be the local Campana space. We equip

$$(X_m, \Delta_H)(\mathbf{A}_k) = \prod_v{}' (X_m(k_v), \mathcal{C}_v(\mathscr{X}_m, \Delta_H))$$

with the restricted-product topology, which we call the *Campana topology*.

Theorem 1.1. *Let $a, b, c \in k^\times$, let $m \in k\setminus\{0, 4abc\}$, and let X_m be the surface in (1.1). For every hyperplane section H of X_m and every integer $n \ge 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X_m, (1-1/n)H)$; that is,*

$$\overline{(X_m \setminus H)(k)} = (X_m, \Delta_H)(\mathbf{A}_k)^{\mathrm{Br}(X_m)}.$$

The closure is taken in the Campana topology.

Ordinary strong approximation behaves quite differently. Suppose that $k = \mathbf{Q}$ and $a = b = c = 1$, and let U_m be the affine variety defined by (1.2). Weak strong approximation always fails on U_m [CTWX20, Theorem 1.1]; hence strong approximation with the Brauer–Manin obstruction off the infinite place also always fails on U_m [CTWX20, Corollary 1.2]. Thus, Theorem 1.1 does not hold when $n = \infty$.

Any k-line induces a fibration of X_m over $\mathbf{P}^1$. Such fibrations play a key role in the proof of this theorem. Its proof splits into two parts. If $H \ne D$, we may choose two sides of D, which are a pair of intersecting k-lines. Proposition 3.6 therefore settles this case directly by the fibration method.

Let f be the fibration defined by D_x. If $H = D$, then the fibre $f^{-1}(\infty) = D_y + D_z$ lies in the boundary, which may be regarded as having multiplicity n. The corresponding fibration problem is therefore particularly difficult, since it has one multiple fibre and four additional nonsplit fibres. Instead, we use the action of Vieta involutions, which allows us to move any local point with reduction on D arbitrarily close to the vertices of D; this observation allows us to complete the proof. The three vertices of D are essential to the argument,

so this method no longer applies if they are removed from X_m, although one may expect similar approximation properties after removing them, by arithmetic purity.

In fact, our result applies more broadly. For a smooth cubic surface containing two distinct k-lines, Campana strong approximation with the Brauer–Manin obstruction holds if H does not contain both of these lines; see Proposition 3.6. If the two lines are disjoint, then H automatically fails to contain both. Hence, Campana strong approximation holds for every H. Here the Brauer–Manin obstruction vanishes, since the surface is k-rational. For a smooth cubic surface containing a k-line ℓ, we also prove the same result if H avoids the nodes of all singular fibres and $\ell \not\subset H$; see Proposition 3.5. This shows that the approximation property holds for any H in general position.

Once Theorem 1.1 has been established, its equal-weight boundary case yields the following unequal-weight corollary.

Corollary 1.2. *For any integers $n_x, n_y, n_z \geq 2$, let*

$$\Delta_{\boldsymbol{n}} = \left(1 - \frac{1}{n_x}\right) D_x + \left(1 - \frac{1}{n_y}\right) D_y + \left(1 - \frac{1}{n_z}\right) D_z. \tag{1.3}$$

Campana strong approximation with the Brauer–Manin obstruction holds for $(X_m, \Delta_{\boldsymbol{n}})$.

The paper is organised as follows. In Section 2 we define the Campana topology and Campana strong approximation, and we also establish some basic properties of this topology. In Section 3 we establish a two-conic-bundle criterion by the fibration method. Applying it to cubic surfaces, we treat the hyperplane sections of X_m other than D. Section 4 treats the hardest case, that of the boundary triangle, using Vieta involutions.

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2. Notation and preliminaries

2.1. **Campana topology and strong approximation.** Let k be a number field with a fixed algebraic closure $\overline{k}$. Denote by Ω_k the set of all places of k, by $\Omega_k^\infty \subset \Omega_k$ its archimedean places, and by $\mathbb{A}_k$ its adèle ring. For each place v, let k_v be the completion of k at v. When v is finite, let $\mathcal{O}_v$ be its valuation ring with the residue field $\mathbb{F}_v$, π_v a uniformizer, and ord_v the normalized valuation. We set $\mathrm{ord}_v(0) = \infty$. The finite set $S \subset \Omega_k$ is always assumed to contain Ω_k^∞, and we write $\mathcal{O}_S$ for the ring of S-integers.

Given a k-variety Y, we write $\overline{Y} = Y \times_k \overline{k}$. Its Brauer group is $\mathrm{Br}(Y) = H^2_{\text{ét}}(Y, \mathbf{G}_{\mathrm{m}})$, and its algebraic Brauer group is $\mathrm{Br}_1(Y) = \ker(\mathrm{Br}(Y) \longrightarrow \mathrm{Br}(\overline{Y}))$. Let $\mathrm{Br}_0(Y) \subseteq \mathrm{Br}_1(Y)$ be the constant Brauer subgroup.

For a smooth k-variety Y and a subgroup $B \subseteq \mathrm{Br}(Y)$, set

$$Y(\mathbf{A}_k)^B = \left\{ (P_v) \in Y(\mathbf{A}_k) : \sum_{v \in \Omega_k} \mathrm{inv}_v\, \alpha(P_v) = 0 \text{ for every } \alpha \in B \right\}.$$

Here $Y(\mathbf{A}_k)$ carries its usual adelic topology; if Y is proper, this topology coincides with the product topology. Sometimes we abbreviate $Y(\mathbf{A}_k)^{\mathrm{Br}(Y)}$ as $Y(\mathbf{A}_k)^{\mathrm{Br}}$.

Let X be a smooth, separated, geometrically integral k-variety, and let

$$\Delta = \sum_{i\in I}\left(1-\frac{1}{n_i}\right)B_i, \qquad n_i \in \mathbf{Z}_{\geq 2}\cup\{\infty\},$$

where the B_i are distinct prime divisors of X and we set $1/\infty = 0$.

Let $\mathscr{X}$ be a separated flat $\mathcal{O}_S$-model of X, let $\mathscr{B}_i$ be the Zariski closure of B_i in $\mathscr{X}$, and let

$$\mathscr{D} = \sum_{i\in I}\left(1-\frac{1}{n_i}\right)\mathscr{B}_i.$$

Then $(\mathscr{X},\mathscr{D})$ is called an $\mathcal{O}_S$-model of (X,Δ). Here X is not assumed to be proper.

For any local integral point $x\in\mathscr{X}(\mathcal{O}_v)$, let $\widetilde{x}:\operatorname{Spec}\mathcal{O}_v\to\mathscr{X}_{\mathcal{O}_v}$ denote the corresponding section. For a closed subscheme $\mathscr{E}\subset\mathscr{X}_{\mathcal{O}_v}$ with ideal sheaf $\mathcal{I}_{\mathscr{E}}$, set $J_{\mathscr{E},x}=\widetilde{x}^{-1}\mathcal{I}_{\mathscr{E}}\cdot\mathcal{O}_v$ and define the intersection multiplicity

$$n_v(\mathscr{E},x)=\begin{cases}\operatorname{length}_{\mathcal{O}_v}(\mathcal{O}_v/J_{\mathscr{E},x}), & J_{\mathscr{E},x}\neq 0,\\ \infty, & J_{\mathscr{E},x}=0.\end{cases}$$

Thus, when the multiplicity is finite, $J_{\mathscr{E},x}=(\pi_v^{n_v(\mathscr{E},x)})$. If $\mathscr{E}$ is locally generated by a function f near x, then we have $n_v(\mathscr{E},x)=\operatorname{ord}_v(f(x))$.

For any place $v\notin S$, we write

$$B_i\times_k k_v=\bigcup_{\lambda\in I_i(v)}B_{i,\lambda},$$

where the $B_{i,\lambda}$ are the k_v-irreducible components of B_i. Let $\mathscr{B}_{i,\lambda}$ denote their closures in $\mathscr{X}_{\mathcal{O}_v}$. For $2\leq n<\infty$, put $\mathcal{N}_n=\{0\}\cup\mathbf{Z}_{\geq n}\cup\{\infty\}$, and set $\mathcal{N}_\infty=\{0\}$. Let $B=\sum_{i\in I}B_i$ and let $X^\circ=X\setminus B$. Following the local-component convention of [NS24, Definition 2.4], for any finite S, we set

$$\begin{aligned}&\mathcal{C}_v(\mathscr{X},\Delta)=\{x\in\mathscr{X}(\mathcal{O}_v): n_v(\mathscr{B}_{i,\lambda},x)\in\mathcal{N}_{n_i}\text{ for all } i\in I\text{ and }\lambda\in I_i(v)\}, && v\notin S,\\ &\mathcal{C}_v(\mathscr{X},\Delta)=X(k_v), && v\in S.\end{aligned}$$

These local Campana inequalities are imposed on each k_v-irreducible component separately, unlike the conditions in [PSTVA21, Definition 3.4], which are imposed on the full closure of each prime boundary k-divisor. For any $\mathcal{O}_S$-model $\mathscr{X}$ and finite $T\supseteq S$, put

$$\mathcal{A}_T(\mathscr{X},\Delta)=\prod_{v\in T}X(k_v)\times\prod_{v\notin T}\mathcal{C}_v(\mathscr{X},\Delta)$$

with the product topology. When X and its model are proper, $\mathscr{X}(\mathcal{O}_v)=X(k_v)$, so we recover the local spaces of [NS24].

Define the *Campana adelic space* by

$$(X,\Delta)(\mathbf{A}_k)=\prod_v{}'(X(k_v),\mathcal{C}_v(\mathscr{X},\Delta))=\bigcup_{T\supseteq S\text{ finite}}\mathcal{A}_T(\mathscr{X},\Delta).$$

The restricted-product topology, which we call the *Campana topology*, has a basis consisting of

$$\prod_{v\in F} W_v \times \prod_{v\notin F} \mathcal{C}_v(\mathscr{X},\Delta), \qquad F \supseteq S \text{ finite},$$

where each $W_v \subseteq X(k_v)$ is open. As with the adelic space $X(\mathbf{A}_k)$, the Campana adelic space and its topology do not depend on the choice of an $\mathcal{O}_S$-model, since any two integral models become isomorphic after removing finitely many places. By the Lang–Weil estimate, $\mathscr{X}^\circ(\mathcal{O}_v) \neq \emptyset$ for almost all v. Hence the Campana adelic space is dense in $X(\mathbf{A}_k)$, and its topology is finer than the subspace topology inherited from $X(\mathbf{A}_k)$. If the boundary Δ is empty, it recovers $X(\mathbf{A}_k)$. Every rational point $x \in X(k)$ lies in the Campana adelic space if all weights are finite. In fact, if x lies on the boundary, the intersection multiplicity is either 0 or ∞ outside finitely many places. If $x \in X^\circ(k)$, such an x extends to the model and avoids the boundary at almost all places. Points in X° are called *strict points*.

For any finite $T \subset \Omega_k$, the projection pr^T omits the components at the places in T. The image $\mathrm{pr}^T\big((X,\Delta)(\mathbf{A}_k)\big)$ carries the topology induced by the restricted product

$$\prod_{v\notin T}{}' (X(k_v), \mathcal{C}_v(\mathscr{X},\Delta)).$$

If $\prod_{v\in T} X(k_v) \neq \varnothing$, the projected image agrees with this restricted product. Denote

$$(X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}(X)} = (X,\Delta)(\mathbf{A}_k) \cap X(\mathbf{A}_k)^{\mathrm{Br}(X)}.$$

Definition 2.1. We say that (X,Δ) satisfies:

- *Campana strong approximation* if
$$\overline{X^\circ(k)} = (X,\Delta)(\mathbf{A}_k);$$
- *Campana strong approximation with the Brauer–Manin obstruction* if
$$\overline{X^\circ(k)} = (X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}};$$
- *Campana strong approximation off T* if
$$\overline{\mathrm{pr}^T(X^\circ(k))} = \mathrm{pr}^T((X,\Delta)(\mathbf{A}_k));$$
- *Campana strong approximation with the Brauer–Manin obstruction off T* if
$$\overline{\mathrm{pr}^T(X^\circ(k))} = \mathrm{pr}^T((X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}}),$$

where each closure is taken with respect to the topology on the right-hand side.

Our notion of Campana strong approximation differs from the strong approximation property in [MNS24, Definitions 2.15 and 2.23]. In [MNS24], strong approximation is defined for a fixed model. The restricted-product topology uses reference subsets $\mathscr{U}(\mathcal{O}_v)$ with $\mathscr{U} = \mathscr{X} \setminus \bigcup_i \mathscr{B}_i$. All neighborhoods therefore require avoiding the boundary outside a finite set of places. However, our topology permits positive boundary intersection. In fact, our notion of Campana strong approximation is equivalent to Campana weak approximation for every finite S and every $\mathcal{O}_S$-model; see [NS24, Definition 2.7(ii)]. Finally, the strict

Brauer–Manin condition in [MNS24, Definition 3.7] is defined using $\mathrm{Br}(X^\circ)$, while we use $\mathrm{Br}(X)$.

If all weights are infinite, then $\mathcal{C}_v(\mathscr{X},\Delta) = \mathscr{X}^\circ(\mathcal{O}_v)$ for $v \notin S$. Campana strong approximation for (X,Δ) is equivalent to ordinary strong approximation for X°.

The following statements are contained in [Moe24, Propositions 3.15 and 4.6].

Lemma 2.2. *The space $\mathcal{C}_v(\mathscr{X},\Delta)$ is open and closed in $X(k_v)$. For any dense open subset V of X, $\mathcal{C}_v(\mathscr{X},\Delta)\cap V(k_v)$ is dense in $\mathcal{C}_v(\mathscr{X},\Delta)$.*

Proof. For $v \in S$, the set is simply $X(k_v)$, so no further argument is needed. If $v \notin S$, for any $s \in \mathscr{X}(\mathcal{O}_v)$, the reduction $s \mod \pi_v^n$ is locally constant. For a finite weight n_i, failure of the Campana condition means that the intersection multiplicity lies in $\{1,\dots,n_i-1\}$; for an infinite weight, we require that $\bar{s}$ does not lie in the boundary. Therefore, the local Campana set is open and closed. By the implicit function theorem [Poo17, Proposition 3.5.73], $\mathcal{C}_v(\mathscr{X},\Delta)\cap V(k_v)$ is dense in $\mathcal{C}_v(\mathscr{X},\Delta)$. □

The following comparison is contained in [Moe24, Propositions 4.12 and 4.13].

Proposition 2.3. *Campana strong approximation for (X,Δ) is equivalent to Campana weak approximation for every finite $S \supseteq \Omega_k^\infty$ and every $\mathcal{O}_S$-model of (X,Δ); see* [NS24, Definition 2.7(ii)]. *The same equivalence holds after imposing the Brauer–Manin obstruction throughout.*

Proof. First, suppose that Campana strong approximation holds for (X,Δ). For an $\mathcal{O}_S$-model, $\mathcal{A}_S(\mathscr{X},\Delta)$ is an open subset of $(X,\Delta)(\mathbf{A}_k)$, so the density of $X^\circ(k)$ in $(X,\Delta)(\mathbf{A}_k)$ implies density in $\mathcal{A}_S(\mathscr{X},\Delta)$. Conversely, given an open subset of $(X,\Delta)(\mathbf{A}_k)$, without loss of generality, we may assume that it has the form $\prod_{v\in S} W_v \times \prod_{v\notin S}\mathcal{C}_v(\mathscr{X},\Delta)$ for a given $\mathcal{O}_S$-model. This is an open subset of $\mathcal{A}_S(\mathscr{X},\Delta)$. Hence Campana weak approximation for S and this $\mathcal{O}_S$-model yields a rational point of X° in this open subset, proving Campana strong approximation for (X,Δ). The same argument applies after intersecting with the Brauer–Manin set. □

Let $\Delta_1 \geq \Delta_2$ be Campana divisors on X with weights in $\mathbf{Z}_{\geq 2}\cup\{\infty\}$. For a given $\mathcal{O}_S$-model, the local Campana conditions immediately give

$$\mathcal{C}_v(\mathscr{X},\Delta_1) \subseteq \mathcal{C}_v(\mathscr{X},\Delta_2).$$

The argument in part (1) below appears in [Moe24, Proposition 4.5(1)].

Lemma 2.4. *Let $T \subset \Omega_k$ be finite, possibly empty.*

(1) *Campana strong approximation off T for (X,Δ_1) implies Campana strong approximation off T for (X,Δ_2).*
(2) *Assume that $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ is finite. Campana strong approximation with the Brauer–Manin obstruction off T for (X,Δ_1) implies the corresponding property for (X,Δ_2).*

Proof. The inclusion $(X,\Delta_1)(\mathbf{A}_k) \hookrightarrow (X,\Delta_2)(\mathbf{A}_k)$ is continuous and dense, and the same holds for the spaces away from T. Assertion (1) then follows from continuity.

For (2), we only need to show the density of $(X,\Delta_1)(\mathbf{A}_k)^{\mathrm{Br}}$ in $(X,\Delta_2)(\mathbf{A}_k)^{\mathrm{Br}}$. We fix an integral $\mathcal{O}_k$-model $\mathscr{X}$ of X. Choose a finite subset $B\subset \mathrm{Br}(X)$ that surjects onto $\mathrm{Br}(X)/\mathrm{Br}_0(X)$. Enlarge the finite set S so that the elements of B extend to $\mathrm{Br}(\mathscr{X}_{\mathcal{O}_S})$. Then the evaluation of B at x_v vanishes for every $v\notin S$ and every $x_v\in\mathscr{X}(\mathcal{O}_v)$. For any point $(x_v)\in(X,\Delta_2)(\mathbf{A}_k)^{\mathrm{Br}}$, we can modify x_v at all v outside a sufficiently large finite set so that $x_v\in\mathcal{C}_v(\mathscr{X},\Delta_1)$ and $(x_v)\in X(\mathbf{A}_k)^{\mathrm{Br}}$, i.e., $(x_v)\in(X,\Delta_1)(\mathbf{A}_k)^{\mathrm{Br}}$. This proves the density of $(X,\Delta_1)(\mathbf{A}_k)^{\mathrm{Br}}$ in $(X,\Delta_2)(\mathbf{A}_k)^{\mathrm{Br}}$, completing the proof. □

2.2. **Approximation on rational conics.**

Proposition 2.5. *Let C be a smooth conic with a k-point, and let $E\subset C$ be a reduced effective divisor of degree at most two. For any assignment of finite weights to the closed points of E, the resulting pair satisfies Campana strong approximation.*

Proof. Identify C with $\mathbf{P}^1_k$. If E is empty, the assertion reduces to ordinary weak approximation. Otherwise, a reduced divisor of degree at most two is supported either on at most two k-points or on a single closed point of degree two. Campana weak approximation for every S and every model follows from [NS24, Propositions 3.5 and 3.15] (or from [Wei26, Theorem 1.1]). The claim now follows from Proposition 2.3. □

Lemma 2.6. *Suppose (X,Δ) has finite weights. Let W be an open subset of $(X,\Delta)(\mathbf{A}_k)$. Suppose that there exists a smooth rational conic $C\subset X$, not contained in B, satisfying the following conditions:*

- *$(C\cap B)_{\mathrm{red}}$ has degree at most two;*
- *$C(\mathbf{A}_k)\cap W\neq\emptyset$.*

Then there is a strict rational point of C lying in W.

Proof. To each closed point of $C\cap B$, we assign the maximal weight among the components of B containing it, and we denote the resulting boundary on C by E. It is clear that the natural map $(C,E)(\mathbf{A}_k)\to(X,\Delta)(\mathbf{A}_k)$ is continuous. Hence $C(\mathbf{A}_k)\cap W$ contains a nonempty open subset of $(C,E)(\mathbf{A}_k)$, and the assertion follows from Proposition 2.5. □

3. A criterion for conic bundles and cubic surfaces

Let X be a smooth, proper, geometrically rational surface with a conic fibration $f: X\to\mathbf{P}^1_k$. Assume that every geometrically singular fibre consists of two distinct lines. Let Z_f be the finite reduced scheme of their nodes.

Let H be a reduced effective divisor. Fix a proper $\mathcal{O}_S$-model $\mathscr{X}$ of X, and let $\mathscr{Z}_f$ be the closure of Z_f. We always enlarge S so that:

- the model $\mathscr{X}_{\mathcal{O}_S}$ is smooth over $\mathcal{O}_S$,
- f extends to an $\mathcal{O}_S$-morphism $\widetilde{f}:\mathscr{X}_{\mathcal{O}_S}\to\mathbf{P}^1_{\mathcal{O}_S}$,
- $\mathscr{Z}_f$ is finite étale over $\mathcal{O}_S$,
- the reductions of singular fibres of $\widetilde{f}$ are pairs of distinct lines whose nodes are the reductions of Z_f,
- if H contains no fibre of f, no fibre is contained in the closure of H,

- if $H \cap Z_f = \varnothing$, their reductions do not meet,
- if a split fibre component is not contained in H, then it contains a smooth $\mathbf{F}_v$-point outside H.

Lemma 3.1. *Let H be a reduced effective divisor containing no fibre of f, and assume that its restriction to the generic fibre is reduced of degree at most two. Let C be a smooth k-fibre containing a point $R \in C(k)$. For any $v \notin S$, if*

$$\bar{R}_v \notin \mathscr{Z}_f(\mathbf{F}_v) \qquad or \qquad H \cap Z_f = \varnothing,$$

then there exists $P_v \in C(k_v)$ whose reduction lies outside H.

Proof. The special fibre of the closure of C is either smooth or a pair of distinct lines. In the smooth case, it has a rational point outside H; in the split case, a component not contained in H has a smooth rational point outside H. These points lift to the chosen fibre by Hensel's lemma, applied where f is smooth.

A non-split singular conic over the residue field has only one rational point, namely its node. Hence the reduction of R must be the node. This is impossible when R avoids Z_f. If instead H avoids Z_f, then the section defined by R itself avoids H. This covers all cases. □

Suppose that X has another conic fibration g over $\mathbf{P}^1_k$. Let A be the g-image of Z_f in $\mathbf{P}^1_k$. Enlarge S so that the corresponding properties stated for f also hold for g, the closure $\mathscr{A}$ of A is étale over $\mathcal{O}_S$ and $\widetilde{g}(\mathscr{Z}_f) \subset \mathscr{A}$.

Lemma 3.2. *Suppose that $g : X \to \mathbf{P}^1_k$ is a second conic bundle. Let $A = g(Z_f) \subset \mathbf{P}^1_k$. For every $a \in A$, suppose that the fibre $g^{-1}(a)$ is either smooth or a pair of lines defined over $k(a)$. Then, for any $v \notin S$ and any $R_0 \in X(k)$ lying on a smooth g-fibre C_s with $s \notin A$, there is a point $R \in C_s(k)$ arbitrarily close to R_0 at the places in S, whose reductions avoid Z_f at all $v \notin S$.*

Proof. Let $T = \{v \notin \ \mathrm{S} : \bar{s}_v \in \mathscr{A}\}$, which is a finite set because $s \notin A$. For any $v \in T$, the residue fibre is smooth or split, so we may choose a smooth $\mathbf{F}_v$-point outside $\mathscr{Z}_f$. On the fibre over $\bar{s}$, Hensel's lemma then lifts this point to $Q_v \in C_s(k_v)$.

The smooth conic C_s contains R_0, so $C_s \simeq \mathbf{P}^1_k$. Weak approximation on C_s gives a point $R \in C_s(k)$ arbitrarily close to R_0 at $v \in S$ and sufficiently close to Q_v that $\bar{R} = \bar{Q}_v$ at $v \in T$. If $v \notin S \cup T$, then the reduction $\bar{s}_v$ of s lies outside $\mathscr{A}$. Therefore, every point of $C_s(k_v)$ has reduction outside $\mathscr{Z}_f$, since $\widetilde{g}(\mathscr{Z}_f) \subset \mathscr{A}$. The reduction of R therefore avoids Z_f at every place outside S. □

Theorem 3.3. *Assume that $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$ and that f and g satisfy the hypotheses of Lemma 3.2. Let H be a reduced effective divisor containing no fibre of f. Suppose that the restriction of H to a general f-fibre is of degree at most two. For every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X, (1-1/n)H)$.*

Proof. Let S be the finite set of exceptional places of Lemma 3.2. Let $\Delta = (1-1/n)H$. For any $(P_v) \in (X,\Delta)(\mathbf{A}_k)^{\mathrm{Br}(X)}$ and any open neighbourhood $\prod_{v\in T} W_v \times \prod_{v\notin T} \mathcal{C}_v(\mathscr{X},\Delta)$ of (P_v), where $T \supset S$ is finite, it suffices to find a strict rational point in this neighbourhood.

Since $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$, we may choose a point $R_0 \in X(k)$ lying in W_v for $v \in T$ such that its g-fibre is smooth with parameter outside A and its f-fibre is smooth. Lemma 3.2 gives a point R, arbitrarily close to R_0 at T, whose reductions avoid Z_f outside T. We may assume that the f-fibre C containing R is smooth and that its intersection with H has degree at most two. The conic C contains R and is therefore rational. Lemma 3.1 gives a local point on C whose section avoids H at every $v \notin T$. Applying Lemma 2.6 to C, we obtain a strict Campana point arbitrarily close to R, hence in the given neighbourhood. □

Turning to cubic surfaces, first we have the following approximation property.

Proposition 3.4. *Let $X \subset \mathbf{P}^3_k$ be a smooth cubic surface with a k-line ℓ. The rational points of X are dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$.*

Proof. We contract the k-line ℓ to produce a smooth del Pezzo surface Y of degree four; the image of ℓ is a k-point. By the theorem of Salberger and Skorobogatov [SS91, Theorem 0.1], Y satisfies weak approximation with the Brauer–Manin obstruction. By birational invariance [HS03, p. 1990], the same approximation property holds on X. □

On a smooth cubic surface X, every hyperplane section H is reduced. Over $\overline{k}$, a nonreduced section would be $2M+M'$ or $3M$, where M, M' are lines on X. If $H = 2M+M'$, then

$$H.M = 2M^2 + M'.M = -2 + 1 = -1.$$

If $H = 3M$, then similarly $H.M = 3M^2 = -3$. Both lead to contradictions since the intersection number of a line with any hyperplane section is always 1.

Proposition 3.5. *Let $X \subset \mathbf{P}^3_k$ be a smooth cubic surface with a k-line ℓ, and let f be the conic bundle given by planes through ℓ. Let H be a hyperplane section not containing ℓ.*

If $H \cap Z_f = \varnothing$ or there is a second bundle g satisfying Lemma 3.2, then, for every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $\big(X, (1-1/n)H\big)$.

Proof. Let L be the hyperplane class. Each f-fibre has class $L-\ell$, so $H\cdot(L-\ell) = 3-1 = 2$. Therefore, the restriction of H to a general fibre has degree two. The plane spanned by such a fibre contains ℓ, hence no f-fibre is contained in H.

Let S be the set of exceptional places of Lemma 3.1. For any adelic Campana point (P_v) orthogonal to $\mathrm{Br}(X)$ and any open neighbourhood $\prod_{v\in T} W_v \times \prod_{v\notin T} \mathcal{C}_v(\mathscr{X},(1-1/n)H)$, where $T \supset S$ is finite, Proposition 3.4 gives a rational point R in W_v for every $v \in T$ such that the f-fibre C containing R is smooth and meets H in a divisor of degree two.

If $H \cap Z_f = \varnothing$, Lemma 3.1 supplies local points on C whose sections avoid H at every $v \notin T$. By Lemma 2.6, we may approximate R at the places in T to obtain a strict Campana point. If a second bundle satisfies Lemma 3.2, apply Theorem 3.3 instead. □

Proposition 3.6. *Let $X \subset \mathbf{P}^3_k$ be a smooth cubic surface containing two distinct k-lines ℓ_1, ℓ_2. If a hyperplane section H does not contain ℓ_1, then, for every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X, (1-1/n)H)$.*

In particular, if the lines intersect, this holds for every hyperplane other than the plane containing both lines; if the lines are disjoint, Campana strong approximation holds for every *hyperplane section H, without the Brauer–Manin obstruction.*

Proof. Let L be the hyperplane class, and let f, g be the conic bundles associated with ℓ_1, ℓ_2. Suppose first that the lines are disjoint. Since no plane contains two disjoint lines, we may assume $\ell_1 \not\subset H$. It is clear that

$$(L - \ell_1) \cdot \ell_2 = (L - \ell_2) \cdot \ell_1 = 1,$$

so ℓ_2 and ℓ_1 are sections of f and g, respectively. A section s of g meets every fibre in a smooth point, since $dg \circ ds = \mathrm{id}$. Every singular g-fibre therefore has a smooth point over its residue field, hence its two components are defined over that field. The fibre hypothesis of Lemma 3.2 holds, so Proposition 3.5 yields the assertion. The section of f shows that its generic conic is rational, so $k(X) \simeq k(t)(u)$ and X is k-rational. Consequently, $\mathrm{Br}(X) = \mathrm{Br}_0(X)$ by birational invariance [Poo17, Corollary 6.8.7].

Now suppose that the lines intersect. Write ℓ_3 for the third line in their common plane; it is a k-line because the residual divisor has degree one. Let $a_0 \in \mathbf{P}^1(k)$ satisfy $g^{-1}(a_0) = \ell_1 + \ell_3$, and set $A = g(Z_f)$. We now verify the fibre hypothesis of Lemma 3.2.

Over $\overline{k}$, take $q \in Z_f$. If its f-fibre is $\ell_2 + \ell_3$, then $q \in \ell_3$, so $g(q) = a_0$. Otherwise, write that f-fibre as $L + M$. Both lines are disjoint from ℓ_2, since ℓ_2 belongs to a different f-fibre. The plane containing L and M is the tangent plane T_qX.

If the g-fibre through q is singular, choose a component N passing through q. Every line on X passing through q lies in T_qX. The section cut out by T_qX on X is $\ell_1 + L + M$, so N is one of ℓ_1, L, M. Every component of a g-fibre meets ℓ_2. Thus $N = \ell_1$, since $L + M$ is disjoint from ℓ_2. The g-fibre is $\ell_1 + \ell_3$. Consequently, every fibre of g above A is either smooth or the split fibre $\ell_1 + \ell_3$. This verifies the hypotheses of Lemma 3.2, and the assertion follows from Proposition 3.5. □

Proposition 3.7. *Theorem 1.1 holds for every hyperplane section $H \neq D$.*

Proof. The surface X_m has three k-lines D_x, D_y, D_z, all lying in the hyperplane section D. Since $H \neq D$, at least one line is not contained in H, and the assertion follows from Proposition 3.6. □

4. The hyperplane section $H = D$

The pencil of planes through D_x defines a conic bundle $f : X_m \to \mathbf{P}^1_k$. For finite t, the fibre is

$$C_t : \quad x = tw, \qquad by^2 + cz^2 - tyz + (at^2 - m)w^2 = 0.$$

It is smooth if and only if

$$(t^2 - 4bc)(at^2 - m) \neq 0. \tag{4.1}$$

The fibre at infinity is $f^{-1}(\infty) = D_y + D_z$.

Let $T_0 \supset \Omega_k^\infty$ be a finite set of places containing all finite places at which one of a, b, c is not a unit or m is not integral. We fix the natural projective $\mathcal{O}_{T_0}$-model $\mathscr{X}_m$ of X_m defined by (1.1).

Set the three vertices of D by

$$E = \{[1:0:0:0], [0:1:0:0], [0:0:1:0]\}, \qquad V_m = X_m \setminus E.$$

Let $\mathscr{E}$ be the closure of E in $\mathscr{X}_m$ and let $\mathscr{V} = \mathscr{X}_m \setminus \mathscr{E}$. Recall that $U_m = X_m \setminus D$.

Lemma 4.1. *Let $T \supset T_0$ be a finite set of places and let $R_0 \in U_m(k)$. Then there is a point $R \in U_m(k)$, arbitrarily close to R_0 at every place of T, such that*

$$R \in \mathscr{V}(\mathcal{O}_v) \qquad \text{for every } v \notin T. \tag{4.2}$$

Proof. By Proposition 3.4, we may assume without loss of generality that $t = x(R_0)$ satisfies (4.1), i.e., that the fibre C_t is smooth. Let $\widetilde{f}$ be the extension of f to the natural projective model over $\mathcal{O}_{T_0}$. Let

$$T_1 = T \cup \{v \notin T : \bar{t}_v = \bar{\infty}\},$$

which is a finite set. For every $v \in T_1 \setminus T$, choose $\bar{p}_v = [1:0:1:0]$, a smooth $\mathbf{F}_v$-point of the fibre at infinity outside E. Applying Hensel's lemma for C_t, we may find a local point $p_v \in C_t(k_v)$ reducing to $\bar{p}_v$.

The smooth conic C_t contains R_0, so $C_t \simeq \mathbf{P}^1_k$. Weak approximation gives $R \in C_t(k) \cap U_m(k)$ arbitrarily close to R_0 at T and sufficiently close to p_v at $T_1 \setminus T$. Hence we may assume $\bar{R}_v \notin E$ for $v \in T_1 \setminus T$. For $v \notin T_1$, we have $\bar{t} \neq \bar{\infty}$, so $\bar{R}_v \notin E$ automatically, since the three vertex sections lie in the fibre at infinity. This proves (4.2). □

4.1. **Growth of boundary multiplicities.** For $v \notin T_0$, primitive integral coordinates $[\xi : \eta : \zeta : \omega]$ for $R \in X_m(k_v)$ give the intersection multiplicities

$$\begin{aligned} n_v(\mathscr{D}_x, R) &= \min\{\mathrm{ord}_v(\xi), \mathrm{ord}_v(\omega)\}, \\ n_v(\mathscr{D}_y, R) &= \min\{\mathrm{ord}_v(\eta), \mathrm{ord}_v(\omega)\}, \\ n_v(\mathscr{D}_z, R) &= \min\{\mathrm{ord}_v(\zeta), \mathrm{ord}_v(\omega)\}. \end{aligned} \tag{4.3}$$

The three Vieta involutions of U_m are

$$\begin{aligned} \sigma_x(x, y, z) &= (yz/a - x, y, z), \\ \sigma_y(x, y, z) &= (x, xz/b - y, z), \\ \sigma_z(x, y, z) &= (x, y, xy/c - z). \end{aligned}$$

These polynomial involutions preserve (1.2) by Vieta's formulas; for example,

$$ax(yz/a - x) = by^2 + cz^2 - m.$$

Put

$$\Phi = \sigma_z \circ \sigma_y \circ \sigma_x.$$

For a point $R \in U_m(k_v)$, we may define the pole-order triple refers to its affine coordinates of U_m. The pole order of a coordinate λ is $\max\{0, -\mathrm{ord}_v(\lambda)\}$ and the pole order of 0 is defined to be zero.

Lemma 4.2. *Let $v \notin T_0$ and $R \in U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)$. Then either R has affine integral coordinates, or its pole-order triple is one of*

$$(0,r,r), \qquad (r,0,r), \qquad (r,r,0), \qquad r \geq 1.$$

After one application of Φ, the corresponding triples are

$$(2r,3r,5r), \qquad (r,2r,3r), \qquad (r,r,2r). \tag{4.4}$$

Proof. Choose primitive coordinates $[\xi : \eta : \zeta : \omega]$ for R, with $\omega \neq 0$. If ω is a unit, the affine coordinates are integral. Otherwise, since $R \in \mathscr{V}(\mathcal{O}_v)$, the reduction of R lies on exactly one side of the boundary triangle. If this side is D_x, then η, ζ are units, and

$$\xi\eta\zeta = \omega(a\xi^2 + b\eta^2 + c\zeta^2 - m\omega^2)$$

implies $\mathrm{ord}_v(\xi) \geq \mathrm{ord}_v(\omega)$. Thus the pole orders are $(0,r,r)$ for $r = \mathrm{ord}_v(\omega) \geq 1$. The other two sides give the remaining cases.

Write $R = (x,y,z)$ and put

$$x' = yz/a - x, \qquad y' = x'z/b - y, \qquad z' = x'y'/c - z.$$

Then $\Phi(R) = (x',y',z')$. For $(0,r,r)$, each successive product has strictly larger pole order, giving $(2r,3r,5r)$.

For $(r,0,r)$, the identity $axx' = by^2 + cz^2 - m$ has a unique term cz^2 of least valuation $-2r$. Thus $\mathrm{ord}_v(x') = -r$. The products $x'z$ and $x'y'$ then have pole orders $2r$ and $3r$, giving $(r,2r,3r)$.

For $(r,r,0)$, only by^2 has the least valuation $-2r$ in the same identity, so $\mathrm{ord}_v(x') = -r$. At the point (x',y,z), the unique term of least valuation $-2r$ in the identity

$$byy' = ax'^2 + cz^2 - m$$

is ax'^2, so $\mathrm{ord}_v(y') = -r$. Since z is integral, the equality $z' = x'y'/c - z$ implies that z' has pole order $2r$. This gives $(r,r,2r)$ and proves (4.4). □

Lemma 4.3. *Let $v \notin T_0$. If $R \in U_m(k_v)$ has pole orders $(A,B,A+B)$ with $0 < A \leq B$, then:*

(1) *the boundary multiplicities are*

$$(n_v(\mathscr{D}_x,R), n_v(\mathscr{D}_y,R), n_v(\mathscr{D}_z,R)) = (B,A,0); \tag{4.5}$$

(2) *$\Phi(R)$ has pole orders*

$$(A+2B,\ 2A+3B,\ 3A+5B). \tag{4.6}$$

Proof. Write $R = (x,y,z)$. Multiplying $[x : y : z : 1]$ by π_v^{A+B} gives primitive coordinates with valuations $(B,A,0,A+B)$. Formula (4.3) therefore gives (4.5).

Since $0 < A \leq B$, each successive product has strictly larger pole order. A direct computation proves (4.6). □

Proposition 4.4. *Let $n \geq 2$ be an integer. Let $v \notin T_0$ and let $\mathscr{X}_m$ be the natural projective $\mathcal{O}_{T_0}$-model. Choose $N \geq 1$ such that $3^{N-1} \geq n$, and let $\varphi = \Phi^N$. Then*

$$\varphi\big(U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)\big) \subset \mathcal{C}_v(\mathscr{X}_m, (1-1/n)D).$$

Proof. Let $R \in U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)$. If its affine coordinates are integral, they remain integral under the Vieta involutions, so all boundary multiplicities of $\varphi(R)$ are zero. Otherwise, Lemma 4.2 gives pole orders $(A, B, A+B)$ after the first application of Φ, with $1 \le A \le B$. Lemma 4.3 preserves this form and at least triples the smallest entry with each further application of Φ. After N applications, this entry is therefore at least $3^{N-1}A \ge n$. By (4.5), the two nonzero boundary multiplicities are then at least n, proving the Campana conditions. □

4.2. **Proof of the theorem.**

Proof of Theorem 1.1. For $H \neq D$, apply Proposition 3.7. Now assume that $H = D$. Let T_0 and the $\mathcal{O}_{T_0}$-model $\mathscr{X}_m$ be as above.

Fix $\varphi = \Phi^N$ as in Proposition 4.4. For an adelic Campana point (P_v) orthogonal to $\mathrm{Br}(X_m)$, choose an arbitrary open neighbourhood $\prod_{v\in T} \mathcal{N}_v \times \prod_{v\notin T} \mathcal{C}_v(\mathscr{X}_m, (1-1/n)D)$ of (P_v) for some $T \supset T_0$. Proposition 3.4 gives a rational point A such that $A \in \mathcal{N}_v$ at every $v \in T$ and $R_0 = \varphi^{-1}(A) \in U_m(k)$.

By the continuity of φ, we may choose neighbourhoods of R_0 at the places in T such that their images lie in $\mathcal{N}_v$. Lemma 4.1 gives a point $R \in U_m(k)$ lying in these neighbourhoods such that $R \in \mathscr{V}(\mathcal{O}_v)$ for every $v \notin T$. Put $P = \varphi(R)$. At each $v \in T$, P lies in $\mathcal{N}_v$. Proposition 4.4 gives the remaining Campana conditions for $v \notin T$, which completes the proof. □

Proof of Corollary 1.2. For $M = \max\{n_x, n_y, n_z\}$, one has $(1-1/M)D \ge \Delta_{\boldsymbol{n}}$. Since $\mathrm{Br}(X_m)/\mathrm{Br}_0(X_m)$ is finite, the corollary then follows from Theorem 1.1 with weight M and Lemma 2.4 (2). □

Remark 4.5. The approximation theorem for generalized Markoff surfaces proved above concerns finite weights. Both Vieta iterations and the rational-conic arguments are similarly restricted to finite weights. These methods do not give approximation results for infinite weights. Indeed, the failures of integral approximation recalled in the introduction show that such an extension is impossible in general.

References

[BGS16] J. Bourgain, A. Gamburd, and P. Sarnak, *Markoff triples and strong approximation*, C. R. Math. Acad. Sci. Paris **354** (2016), no. 2, 131–135.

[BN23] M. Bright and R. Newton, *Evaluating the wild Brauer group*, Invent. Math. **234** (2023), 819–891.

[BY21] T. D. Browning and S. Yamagishi, *Arithmetic of higher-dimensional orbifolds and a mixed Waring problem*, Math. Z. **299** (2021), no. 1–2, 1071–1101.

[Cam05] F. Campana, *Fibres multiples sur les surfaces: aspects géométriques, hyperboliques et arithmétiques*, Manuscripta Math. **117** (2005), no. 4, 429–461.

[CL09] S. Cantat and F. Loray, *Dynamics on character varieties and Malgrange irreducibility of Painlevé VI equation*, Ann. Inst. Fourier (Grenoble) **59** (2009), no. 7, 2927–2978.

[Che21] S. Chen, *Integral points on twisted Markoff surfaces*, J. Number Theory **220** (2021), 212–234.

[CTWX20] J.-L. Colliot-Thélène, D. Wei, and F. Xu, *Brauer–Manin obstruction for Markoff surfaces*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **21** (2020), 1257–1313.

[Dao24] Q.-D. Dao, *Brauer–Manin obstruction for integral points on Markoff-type cubic surfaces*, J. Number Theory **254** (2024), 65–102.

[GMR19] A. Gamburd, M. Magee, and R. Ronan, *An asymptotic formula for integer points on Markoff-Hurwitz varieties*, Ann. of Math. (2) **190** (2019), no. 3, 751–809.

[GS22] A. Ghosh and P. Sarnak, *Integral points on Markoff type cubic surfaces*, Invent. Math. **229** (2022), no. 2, 689–749.

[HS03] D. Harari and A. N. Skorobogatov, *The Brauer group of torsors and its arithmetic applications*, Ann. Inst. Fourier (Grenoble) **53** (2003), no. 7, 1987–2019.

[Hur07] A. Hurwitz, *Über eine Aufgabe der unbestimmten Analysis*, Arch. Math. Phys. (3) **11** (1907), 185–196.

[KSS24] S. Kim, D. Schindler, and J. Sivaraman, *On Markoff type surfaces over number fields and the arithmetic of Markoff numbers*, in Research Directions in Number Theory, Assoc. Women Math. Ser. **33**, Springer, 2024, 271–293.

[LM21] D. Loughran and V. Mitankin, *Integral Hasse principle and strong approximation for Markoff surfaces*, Int. Math. Res. Not. IMRN (2021), no. 18, 14086–14122.

[Mar79] A. Markoff, *Sur les formes quadratiques binaires indéfinies*, Math. Ann. **15** (1879), 381–406.

[MNS24] V. Mitankin, M. Nakahara, and S. Streeter, *Semi-integral Brauer–Manin obstruction and quadric orbifold pairs*, Trans. Amer. Math. Soc. **377** (2024), no. 6, 4435–4480.

[MU25] V. Mitankin and J. Uhlemann, *Local-global principles for semi-integral points on Markoff orbifold pairs*, J. Lond. Math. Soc. (2) **112** (2025), no. 6, e70363, 24 pp.

[Moe24] B. Moerman, *Generalized Campana points and adelic approximation on toric varieties*, preprint (2024), arXiv:2407.03048.

[NS24] M. Nakahara and S. Streeter, *Weak approximation and the Hilbert property for Campana points*, Michigan Math. J. **74** (2024), no. 2, 227–252.

[PS24] M. Pieropan and D. Schindler, *Hyperbola method on toric varieties*, J. Éc. polytech. Math. **11** (2024), 107–157.

[PSTVA21] M. Pieropan, A. Smeets, S. Tanimoto, and A. Várilly-Alvarado, *Campana points of bounded height on vector group compactifications*, Proc. Lond. Math. Soc. (3) **123** (2021), no. 1, 57–101.

[Poo17] B. Poonen, *Rational points on varieties*, Graduate Studies in Mathematics **186**, American Mathematical Society, Providence, RI, 2017.

[SS91] P. Salberger and A. N. Skorobogatov, *Weak approximation for surfaces defined by two quadratic forms*, Duke Math. J. **63** (1991), no. 2, 517–536.

[Wei26] D. Wei, *Weak approximation for Campana points on projective spaces*, preprint (2026), arXiv:2610.05032.

Academy of Mathematics and System Science, CAS, Beijing 100190, P. R. China *and* School of Mathematical Sciences, University of CAS, Beijing 100049, P. R. China

Email address: dshwei@amss.ac.cn